

Machine Learning-Based Prediction of Wine Quality Using Physicochemical Properties:

A Comparative Study of Ensemble Learning Models with SHAP Explainability

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GitHub Repository

Repository Information

GitHub Repository

<https://github.com/MBG0903/QM640-Wine-Quality-Capstone>

Notebook

https://github.com/MBG0903/QM640-Wine-Quality-Capstone/blob/main/notebooks/QM640_Data_Analytics_Capstone_Project_Mahesh_Govind_araj_Interim_Final_Notebook.ipynb

Project Dataset

UCI Machine Learning Repository – Wine Quality Dataset

<https://archive.ics.uci.edu/dataset/186/wine+quality>

1. Introduction

1.1 Background and Context

Wine quality is a critical factor influencing consumer satisfaction, brand reputation, market competitiveness, and regulatory compliance within the wine industry. Traditional quality assessment relies on sensory evaluation conducted by expert tasters. Although effective, this approach is subjective, time consuming, costly, and difficult to scale for large volume production (Cortez et al., 2009).

Advances in machine learning have enabled objective quality prediction using measurable physicochemical properties such as alcohol content, acidity, sulphates, pH, and sulfur dioxide. By identifying complex relationships among these variables, machine learning models can support consistent quality assessment and informed production decisions, complementing traditional sensory evaluation (Hastie et al., 2021; James et al., 2021).

This study develops an explainable machine learning framework for wine quality prediction using the UCI Wine Quality dataset. Five regression models Linear Regression, Decision Tree Regression, Random Forest Regression, Gradient Boosting Regression, and XGBoost Regression are developed and compared using independent testing, five-fold cross validation, and hyperparameter optimization. The selected model is interpreted using SHapley Additive exPlanations (SHAP) to identify the physicochemical properties driving wine quality predictions, supporting transparent and reliable decision making (Breiman, 2001; Chen & Guestrin, 2016; Lundberg & Lee, 2017).

1.2 Problem Statement

Maintaining consistent wine quality is essential for ensuring customer satisfaction, regulatory compliance, and brand competitiveness. Traditional sensory evaluation is subjective, time consuming, and difficult to scale for large production environments. Although machine learning has demonstrated promising results in wine quality prediction,

many studies focus primarily on predictive accuracy while providing limited comparison of regression models and insufficient transparency in model predictions (Cortez et al., 2009; Chen & Guestrin, 2016). This study addresses these challenges by developing an explainable machine learning based decision support framework that compares multiple regression models and applies SHAP to provide transparent, interpretable, and data driven support for wine quality assessment (Lundberg & Lee, 2017).

1.3 Purpose of the Study

The purpose of this study is to develop an explainable machine learning based decision support framework for predicting wine quality using physicochemical properties. Five regression models are developed and compared using MAE, RMSE, R^2 , five-fold cross validation, and hyperparameter optimization to identify the most robust predictive model. SHapley Additive exPlanations (SHAP) are subsequently applied to explain model predictions and identify the key physicochemical properties influencing wine quality, thereby supporting transparent and data driven quality assessment (Breiman, 2001; Chen & Guestrin, 2016; Lundberg & Lee, 2017).

1.4 Interim Project Status (Progress Snapshot)

To provide an overview of the current progress of the capstone project, the major activities completed, ongoing tasks, and remaining deliverables are summarized in **Table 1**. This progress assessment demonstrates the extent to which the project objectives have been achieved at the interim stage and highlights the activities planned before the final submission.

Table 1

Project Progress Summary

Status	Activities
Completed	• Completed literature review and established the research framework.

Status	Activities
	• Collected, validated, and pre-processed the UCI Wine Quality dataset. • Performed exploratory data analysis, feature analysis, and data quality assessment. • Developed, optimized, and evaluated five machine learning regression models using cross validation and hyperparameter tuning. • Selected the final predictive model and completed SHAP based explainability analysis. • Addressed all research questions and documented the key findings.
In Progress	• Reviewing the report for consistency with the implementation notebook. • Verifying APA 7 formatting, tables, figures, citations, and references. • Refining the GitHub repository and supporting documentation.
Pending	• Submission of the revised interim report. • Preparation and submission of the final capstone report and presentation.

As shown in **Table 1**, the core analytical components of the study, including data preparation, exploratory analysis, feature analysis, model development, hyperparameter optimization, and SHAP explainability, have been completed. The remaining activities primarily focus on refining the report, verifying APA 7 compliance, updating the GitHub repository, and preparing the final capstone submission.

2 Scope and Objectives

2.1 Scope of the Study

This study focuses on developing an explainable machine learning based decision support framework to predict the quality of **Portuguese Vinho Verde red wines** using physicochemical properties obtained from the UCI Machine Learning Repository. The red wine dataset was selected because it is a well-established benchmark dataset containing high

quality laboratory measurements and expert quality ratings, making it suitable for developing and evaluating predictive models (Cortez et al., 2009). The study compares multiple regression models to identify the most accurate, robust, and interpretable solution for wine quality assessment. The findings are intended to support data driven quality control and decision making while demonstrating the practical application of Explainable Artificial Intelligence (SHAP) in predictive analytics (Lundberg & Lee, 2017).

2.2 Research Objectives

The objectives of this study are to:

1. Develop an explainable machine learning based decision support framework for assessing wine quality using physicochemical properties.
2. Compare the predictive performance of multiple regression models to identify the most accurate, robust, and interpretable solution for wine quality assessment.
3. Evaluate model performance using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), coefficient of determination (R^2), and five-fold cross validation.
4. Improve model generalization and predictive performance through hyperparameter optimization.
5. Apply SHapley Additive exPlanations (SHAP) to identify the physicochemical properties that most influence wine quality and enhance the transparency of model predictions.

2.3 Research Questions

RQ1. Which physicochemical properties most influence wine quality?

RQ2. How accurately can machine learning regression models predict wine quality using physicochemical properties?

RQ3. Which machine learning regression model provides the best predictive performance for wine quality prediction?

RQ4. Which physicochemical properties contribute the most to the predictive performance of the best performing machine learning model?

The research questions were formulated based on the approved research proposal. For completeness and traceability, the corresponding research hypotheses are provided in **Appendix B**

3 Literature Survey

3.1 Literature Review Approach

The literature review was conducted to identify research related to wine quality prediction, machine learning regression, ensemble learning, and Explainable Artificial Intelligence (XAI). Peer reviewed journal articles, conference papers, textbooks, and publicly available datasets were identified through Google Scholar, SpringerLink, the ACM Digital Library, the Journal of Machine Learning Research (JMLR), and the UCI Machine Learning Repository. Search keywords included *wine quality prediction*, *machine learning regression*, *Random Forest*, *Gradient Boosting*, *XGBoost*, *SHAP*, *Explainable AI*, and *wine quality dataset*.

Sources were selected based on their relevance to supervised machine learning, regression modelling, explainability techniques, and predictive analytics. The reviewed literature provides the theoretical foundation for the selected algorithms, evaluation methodology, and explainability framework adopted in this study, while supporting the formulation of the research questions and overall research design.

Existing studies have demonstrated that machine learning can predict wine quality using physicochemical properties. However, relatively few studies provide a systematic comparison of multiple ensemble learning models while integrating Explainable Artificial Intelligence techniques such as SHAP to improve model transparency. This study addresses

this gap by combining comparative model evaluation with explainability, thereby providing a decision support framework that is both accurate and interpretable.

To summarize the key findings from the reviewed literature, Table 2 presents the major research themes and the research gap addressed in this study.

Table 2

Summary of Literature Review Findings

Research Area	Key Finding
Wine Quality Prediction	ML improves objective quality assessment
Ensemble Learning	Ensemble models generally outperform single models
Explainable AI	SHAP improves transparency and interpretability
Research Gap	Limited studies integrate comparative modelling with SHAP

As summarized in **Table 2**, the reviewed literature demonstrates the effectiveness of machine learning and ensemble learning for wine quality prediction while highlighting the growing importance of Explainable Artificial Intelligence. The identified research gap provided the motivation for this study, which combines comparative model evaluation with SHAP explainability to develop a transparent and interpretable decision support framework for wine quality assessment.

4 Dataset Description and Sources

4.1 Data Description

This study uses the **Wine Quality Dataset** obtained from the UCI Machine Learning Repository, which contains laboratory measured physicochemical properties and corresponding quality ratings for Portuguese Vinho Verde red wines. The dataset is a widely used benchmark for supervised machine learning because of its public availability, high data

quality, and extensive use in predictive analytics research (Cortez et al., 2009). It contains **1,599 wine samples** and **12 variables**, including **11 physicochemical predictor variables** and one target variable (wine quality). After removing duplicate records, **1,359 unique observations** were retained for model development and evaluation.

4.2 Data Source(s) and Access

Primary Dataset: UCI Machine Learning Repository. (n.d.). *Wine Quality Dataset*.

University of California, Irvine. <https://archive.ics.uci.edu/dataset/186/wine+quality>

Project Repository: GitHub Repository <https://github.com/MBG0903/QM640-Wine-Quality-Capstone>

Table 3

Dataset Overview

Attribute	Description
Dataset Name	Wine Quality Dataset (Red Wine)
Source	UCI Machine Learning Repository
Original Records	1,599
Records After Preprocessing	1,359
Predictor Variables	11
Target Variable	Wine Quality
Total Variables	12
Unit of Analysis	Individual wine sample
Time Period	Not specified by the data source

Table 3 summarizes the key characteristics of the dataset used in this study.

4.3 Data Dictionary

The complete data dictionary is provided in **Appendix A** (Table 4)

4.4 GitHub Data Availability Statement

The raw dataset, processed dataset, implementation notebook, and supporting project files are publicly available through the project GitHub repository to promote transparency and reproducibility.

Raw data: <https://github.com/MBG0903/QM640-Wine-Quality-Capstone/tree/main/data/raw>

Processed data: <https://github.com/MBG0903/QM640-Wine-Quality-Capstone/tree/main/data/processed>

Implementation notebook: https://github.com/MBG0903/QM640-Wine-Quality-Capstone/blob/main/notebooks/QM640_Data_Analytics_Capstone_Project_Mahesh_Govind_araj_Interim_Final_Notebook.ipynb

5 Methodology Workflow

Figure 1 illustrates the end to end methodology adopted in this study. The workflow begins with dataset collection and data preprocessing, followed by exploratory data analysis and feature analysis. Multiple regression models are then developed, optimized, and evaluated before selecting the best performing model. Finally, SHAP explainability is applied to interpret model predictions and identify the key physicochemical factors influencing wine quality, enabling transparent and data driven decision making. A fixed random state (42) was used throughout the analysis to ensure reproducibility of the experimental results.

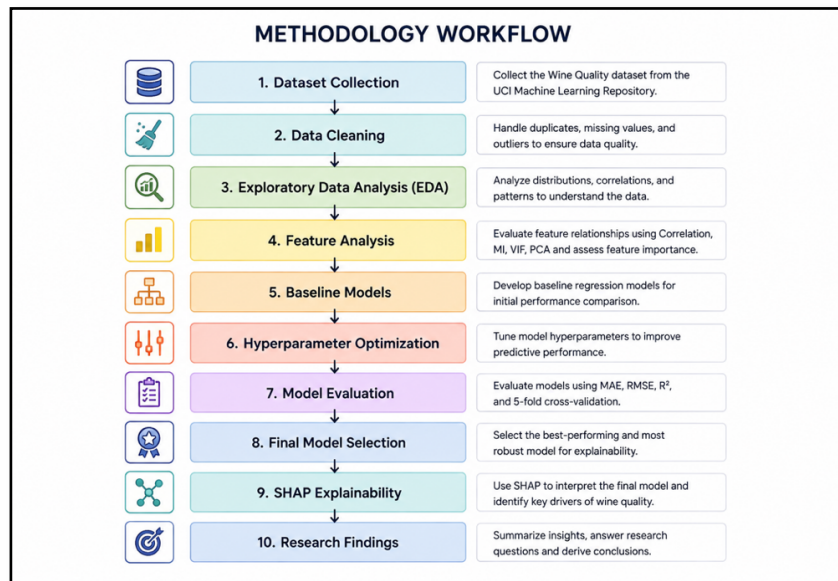


Figure 1

Research Methodology Workflow.

6 Research Findings and Business Implications

This section presents the analytical approach adopted to develop an explainable machine learning based decision support framework for wine quality assessment. It summarizes the key findings from data preprocessing, exploratory data analysis, feature analysis, model development, evaluation, and explainability, providing the evidence used to answer the research questions and select the most suitable predictive model.

6.1 Data Cleaning

Data cleaning was performed to ensure the quality and reliability of the dataset before model development. The assessment included missing values, duplicate records, data types, and outliers. Duplicate records were removed to improve data integrity, while valid outliers were retained to preserve the natural variability of wine characteristics and maintain the authenticity of the production data.

Table 4 summarizes the data cleaning activities performed before model development.

Table 4*Data Cleaning Summary*

Data Quality Check	Result
Original Records	1,599
Missing Values	None
Duplicate Records Removed	240
Final Records	1,359
Data Type Issues	None
Outlier Treatment	Retained

The dataset required minimal preprocessing and was suitable for machine learning analysis after removing duplicate records.

Interpretation

The dataset required minimal preprocessing and was suitable for machine learning analysis. The preprocessing steps improved data reliability while preserving the natural characteristics of the wine samples, providing a robust foundation for predictive modelling.

The dataset was divided into training (80%) and testing (20%) subsets using a fixed random state to ensure reproducibility. All exploratory analysis and preprocessing decisions were performed using the training data before final model evaluation to minimize the risk of data leakage.

6.2 Minimum Sample Size Assessment

To ensure that the dataset was adequate for developing reliable machine learning models, a minimum sample size assessment was performed. The study used **11 predictor variables** and one target variable. Following the guideline proposed by Tabachnick and

Fidell (2019), the recommended minimum sample size for multiple regression is calculated as:

$$N \geq 50 + 8m$$

where N is the minimum required sample size and m is the number of predictor variables.

For this study:

$$N = 50 + (8 \times 11) = 138$$

After duplicate removal, the final dataset contained **1,359 unique observations**, which substantially exceeds the recommended minimum sample size of **138 observations**. Therefore, the available dataset is considered adequate for exploratory data analysis, machine learning model development, cross-validation, hyperparameter optimization, and comparative evaluation of the regression models. The same dataset was used to address all four research questions; therefore, the sample size was considered sufficient for the analyses performed throughout this study (Tabachnick & Fidell, 2019).

6.3 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the distribution of wine quality scores, identify relationships among physicochemical properties, and uncover patterns that influence wine quality. These insights provided a better understanding of the dataset and supported informed feature selection, model development, and business interpretation. Figures 2 and 3 summarize the key findings from the exploratory analysis.

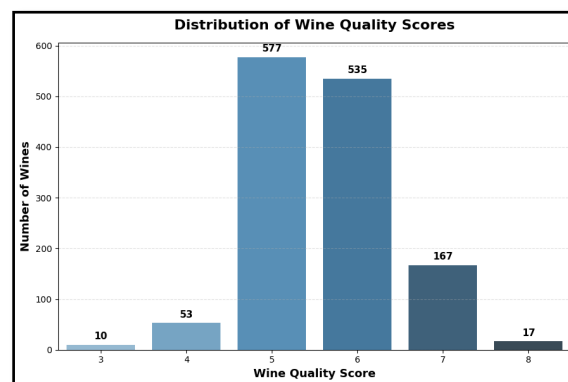


Figure 2*Wine Quality Distribution***Interpretation**

Figure 2 shows that the wine quality distribution is moderately imbalanced, with most samples receiving quality scores of 5 or 6, while very low and very high quality wines occur less frequently. This distribution reflects typical production conditions and indicates that the predictive framework is primarily designed to distinguish variations among average quality wines, enabling wineries to better understand the factors influencing quality consistency and identify opportunities for process improvement.

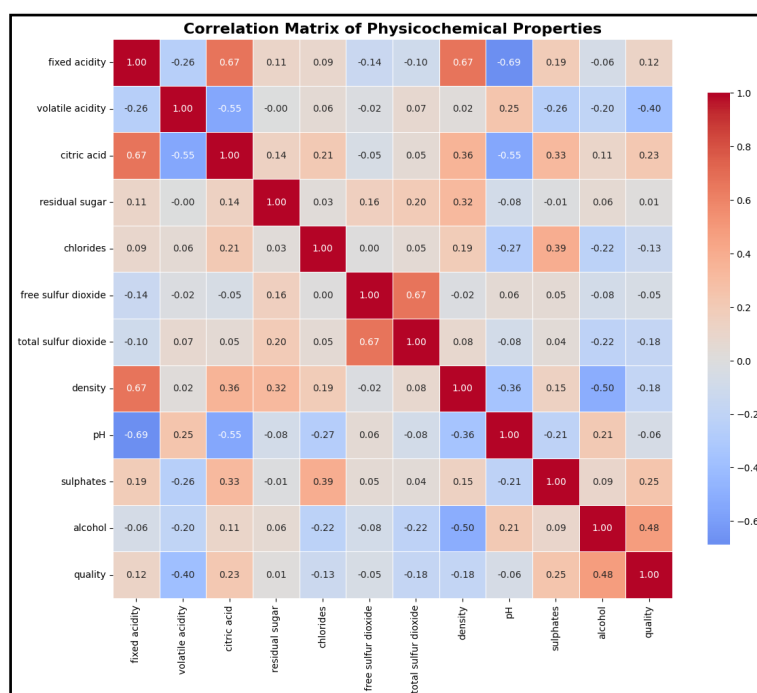
**Figure 3***Correlation Heatmap***Interpretation**

Figure 3 shows that **alcohol** has the strongest positive correlation with wine quality, while **volatile acidity** exhibits the strongest negative correlation. Most of the remaining physicochemical properties demonstrate weak to moderate relationships, indicating that wine

quality is influenced by the combined effect of multiple characteristics rather than a single factor. These findings highlight the importance of adopting a multivariate, data driven approach to support quality assessment and help wineries identify the key factors that influence product consistency.

6.4 Key Findings from Exploratory Data Analysis

Table 5 summarizes the key findings from the exploratory data analysis.

Table 5

Key Findings from Exploratory Data Analysis

Observation	Key Finding
Dataset Distribution	Majority of wine samples have quality scores of 5 or 6.
Missing Values	No missing values detected.
Duplicate Records	240 duplicate records removed during preprocessing.
Strong Positive Correlation	Alcohol with wine quality.
Strong Negative Correlation	Volatile acidity with wine quality.
Feature Relationships	Most predictors exhibit weak to moderate correlations, supporting the use of machine learning models.

The exploratory analysis confirmed that the dataset was suitable for predictive modelling and provided valuable insights for subsequent feature analysis and model development.

Business Insight

The exploratory analysis confirmed that the dataset is suitable for predictive modelling and provided valuable insights into the physicochemical factors influencing wine quality. The observed relationships demonstrate that wine quality is determined by the combined effect of multiple characteristics rather than a single variable. These findings

support the development of an explainable machine learning based decision support framework to assist wineries in improving quality consistency and making more informed production decisions.

6.5 Feature Analysis

Feature analysis was conducted to evaluate the relevance, independence, and predictive contribution of the physicochemical variables before model development. Correlation Analysis, Mutual Information, Variance Inflation Factor (VIF), and Principal Component Analysis (PCA) were performed to understand feature relationships, assess multicollinearity, and validate the suitability of the original predictor variables. The findings confirmed that all eleven physicochemical properties contributed meaningful information and were retained for model development.

Table 6 summarizes the findings from the feature analysis conducted before model development.

Table 6

Feature Analysis Summary

Analysis	Key Finding	Business Value
Correlation Analysis	Alcohol showed the strongest positive correlation with wine quality, while volatile acidity showed the strongest negative correlation.	Identified the key physicochemical properties influencing wine quality.
Mutual Information	Alcohol, sulphates, and volatile acidity exhibited the highest information gain.	Confirmed the predictive importance of the most influential features.

Analysis	Key Finding	Business Value
Variance Inflation Factor (VIF)	No severe multicollinearity was observed among the predictor variables.	Ensured reliable model development by minimizing feature redundancy.
Principal Component Analysis (PCA)	Approximately 95% of the dataset variance was explained by the first nine principal components.	Confirmed that dimensionality reduction was unnecessary and that retaining the original variables improved model interpretability.

The feature analysis confirmed that all predictor variables contributed meaningful information and were retained for model development.

Business Insight

The feature analysis demonstrated that each physicochemical property contributed valuable information for predicting wine quality. Retaining all original variables preserved domain interpretability while enabling SHAP to provide meaningful explanations of feature contributions. This supports a transparent decision support framework that can assist wineries in understanding the factors influencing product quality and improving quality control decisions.

7 Model Development and Evaluation

7.1 Choice of Models with Justification

Five supervised machine learning regression models were evaluated to develop an explainable decision support framework for wine quality assessment. Linear Regression was selected as the baseline model, while Decision Tree Regression, Random Forest Regression, Gradient Boosting Regression, and XGBoost Regression were chosen to evaluate

progressively more advanced modelling techniques. This comparative approach enables the identification of a predictive model that balances accuracy, robustness, and interpretability, thereby supporting reliable and transparent decision making in wine quality assessment (Breiman, 2001; Friedman, 2001; Chen & Guestrin, 2016).

Table 7 summarizes the regression models evaluated and the rationale for their inclusion in this study.

Table 7

Regression Models Evaluated

Model	Purpose / Justification
Linear Regression	Baseline model for comparison
Decision Tree Regression	Captures non-linear relationships and provides interpretability
Random Forest Regression	Reduces overfitting through ensemble learning
Gradient Boosting Regression	Improves predictive performance by sequential learning
XGBoost Regression	Optimized boosting algorithm with strong predictive capability

The selected models provide a balanced comparison of baseline, tree based, and ensemble learning techniques for wine quality prediction.

Business Insight

Evaluating multiple regression models ensured that the selected solution was not only accurate but also robust and suitable for practical application. This comparative approach supports the selection of a predictive framework that can deliver consistent and reliable

quality assessments while maintaining sufficient transparency for business users and quality control professionals.

7.2 Features Included and Feature Engineering

All 11 physicochemical properties were retained as predictor variables for model development. Feature analysis using Correlation Analysis, Mutual Information, Variance Inflation Factor (VIF), and Principal Component Analysis (PCA) confirmed that each variable contributed meaningful predictive information and that no severe multicollinearity existed among the features. Since the original laboratory measured variables adequately represented the characteristics of the wine samples, no additional feature engineering was required. Retaining the original variables also preserved domain interpretability and enabled meaningful SHAP based explanations of model predictions.

No additional feature engineering was required because the original laboratory measured variables adequately represented the characteristics of the wine samples. Data scaling was applied only where necessary for feature analysis (e.g., PCA), while the machine learning models were trained using the original feature values. The complete list of predictor variables is provided in **Appendix A** (Table 8).

7.3 Evaluation Metrics

Model performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), coefficient of determination (R^2), and five-fold cross validation. These metrics provide a comprehensive assessment of prediction accuracy, model robustness, and generalization capability, enabling the selection of the most reliable model for wine quality assessment (Hastie et al., 2021; James et al., 2021).

Table 9 summarizes the evaluation metrics used to assess the performance of the regression models.

Table 9*Model Evaluation Metrics*

Metric	Business Relevance
Mean Absolute Error (MAE)	Measures the average prediction error, indicating how closely the predicted quality matches the actual quality.
Root Mean Squared Error (RMSE)	Measures prediction error while assigning greater importance to larger errors, supporting the selection of reliable models.
Coefficient of Determination (R^2)	Measures how effectively the model explains variations in wine quality.
Five Fold Cross Validation	Evaluates model stability and generalization across multiple data partitions, ensuring robust performance on unseen data.

The selected metrics provide a comprehensive assessment of prediction accuracy, model robustness, and generalization capability.

Business Insight

A combination of accuracy and generalization metrics was used to ensure that the selected model not only achieved strong predictive performance but also demonstrated consistent and reliable behaviour on unseen data. This approach supports the development of a robust decision support framework that can be confidently applied to wine quality assessment.

7.4 Model Performance Summary

The optimized regression models were compared using Test RMSE, Test R^2 , Cross Validation RMSE, and Cross Validation R^2 to evaluate predictive accuracy, robustness, and

generalization capability. Table 10 summarizes the performance of the optimized candidate models.

Table 10 compares the performance of the optimized regression models using the selected evaluation metrics.

Table 10

Comparison of Optimized Regression Models

Model	Test RMSE	Test R ²	CV RMSE	CV R ²
Tuned Decision Tree	0.6929	0.2843	0.6875	0.2975
Tuned Random Forest	0.6377	0.3937	0.6372	0.3945
Tuned Gradient Boosting	0.6296	0.4090	0.6396	0.3891
Tuned XGBoost	0.6316	0.4053	0.6389	0.3912

As shown in **Table 10**, the Tuned Random Forest model provided the best balance between predictive accuracy and generalization capability and was therefore selected as the final predictive model.

Business Insight

The optimized ensemble models consistently outperformed the Decision Tree model. Although the Tuned Gradient Boosting model achieved the highest Test R² (0.4090) and the lowest Test RMSE (0.6296), the Tuned Random Forest model demonstrated the strongest cross validation performance and was therefore selected as the final predictive model. SHAP was subsequently applied to provide transparent explanations of its predictions.

7.4.1 Actual vs Predicted

To complement the numerical evaluation metrics, the predictive performance of the selected Tuned Random Forest Regression model was visually assessed using an Actual versus Predicted plot.

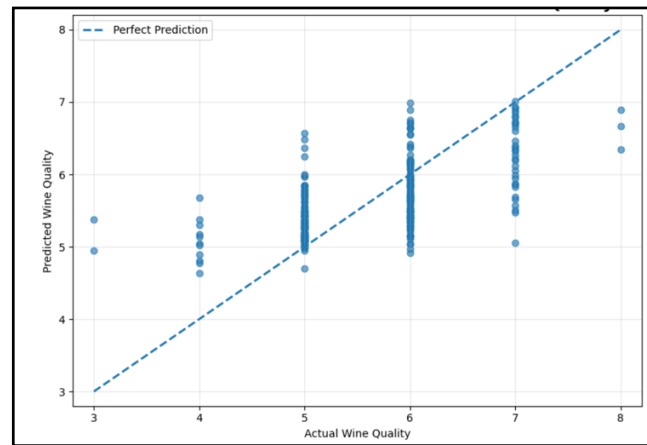


Figure 4

Actual versus Predicted Wine Quality Using the Tuned Random Forest Regression Model.

Most observations are clustered around the reference line, indicating that the selected Tuned Random Forest model provides reasonable predictive accuracy despite the moderate variability inherent in subjective wine quality ratings.

7.5 Preliminary Findings by Research Question

RQ1. Which physicochemical properties most influence wine quality?

Exploratory Data Analysis, feature analysis, and SHAP consistently identified alcohol, sulphates, volatile acidity, and total sulfur dioxide as the most influential physicochemical properties affecting wine quality. Alcohol exhibited the strongest positive relationship with wine quality, whereas volatile acidity showed the strongest negative relationship. These findings provide valuable insights for quality control teams by identifying the key laboratory measurements that should be monitored to improve product consistency.

RQ2. How accurately can machine learning regression models predict wine quality using physicochemical properties?

The evaluated regression models demonstrated moderate predictive capability using only physicochemical properties. Hyperparameter optimization improved the performance of the ensemble learning models, with all optimized ensemble models outperforming the individual Decision Tree Regression model. These results demonstrate the potential of machine learning to support objective and data driven wine quality assessment alongside traditional sensory evaluation.

RQ3. Which machine learning regression model provides the best predictive performance for wine quality prediction?

Although the Tuned Gradient Boosting Regression model achieved the highest Test R^2 (0.4090) and the lowest Test RMSE (0.6296), the Tuned Random Forest Regression model demonstrated the strongest cross validation performance and overall robustness. Consequently, the Tuned Random Forest model was selected as the final predictive model because it provided the best balance between predictive accuracy, generalization capability, and interpretability, making it more suitable for practical decision support.

RQ4. Which physicochemical properties contribute the most to the predictive performance of the best performing machine learning model?

SHAP analysis confirmed that **alcohol, sulphates, volatile acidity, and total sulfur dioxide** were the most influential contributors to the Tuned Random Forest model. The explainability analysis provided transparent insights into how each physicochemical property influenced the predicted wine quality, increasing stakeholder confidence and supporting informed quality improvement decisions.

Table 11 summarizes the key findings of the study in relation to the four research questions.

Table 11*Summary of Research Findings*

Research Question	Key Finding	Business Value
RQ1	Alcohol, sulphates, volatile acidity, and total sulfur dioxide were the most influential predictors.	Enables quality teams to focus on the factors with the greatest impact on wine quality.
RQ2	Ensemble learning models outperformed the Decision Tree model.	Supports objective and consistent quality assessment.
RQ3	Tuned Random Forest provided the best balance of accuracy, robustness, and interpretability.	Suitable for deployment as a decision support framework.
RQ4	SHAP explained how individual physicochemical properties influenced predictions.	Improves transparency and user confidence in AI assisted quality assessment.

As shown in Table 11, the study successfully addressed all four research questions and demonstrated the practical value of combining machine learning with SHAP explainability for transparent wine quality assessment.

7.6 Practical Business Implications

The proposed explainable machine learning based decision support framework has the potential to support wineries by providing objective and data driven quality assessments before sensory evaluation, thereby improving the efficiency and consistency of quality control processes. The identified key quality drivers, particularly alcohol, sulphates, volatile

acidity, and total sulfur dioxide, can help quality assurance teams monitor critical production parameters and take proactive corrective actions to improve product consistency.

Furthermore, the integration of SHAP explainability enhances transparency by providing clear insights into the factors influencing each prediction, increasing stakeholder confidence and supporting informed operational decision making.

7.7 Interim Limitations and Risks

The findings of this study should be interpreted within the scope of the selected dataset and methodology. The analysis was conducted using a single publicly available dataset containing Portuguese Vinho Verde red wines, which may limit the generalizability of the results to other wine varieties and production environments (Cortez et al., 2009). In addition, the predictive models rely solely on physicochemical properties and do not incorporate sensory evaluation, grape variety, production practices, or environmental factors that may also influence wine quality. Although the selected model demonstrated robust predictive performance, further validation using independent industrial datasets would improve confidence in its practical applicability and support broader deployment in winery operations.

7.8 Next Steps

Although the analytical work has been completed, the remaining activities focus on documentation and final project submission.

- Complete the final review of the report and implementation notebook.
- Verify consistency between reported results and implementation outputs.
- Perform the final APA 7 formatting review.
- Finalize the GitHub repository documentation.
- Prepare and submit the final capstone report and presentation.

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Appendix A

Table 4
Data Dictionary

Variable Name	Definition	Data Type	Allowed Values / Range	Missing Value Handling	Notes
Fixed Acidity	Concentration of fixed acids in the wine	Numeric	4.6 to 15.9	No missing values	Predictor
Volatile Acidity	Concentration of volatile acids	Numeric	0.12 to 1.58	No missing values	Predictor
Citric Acid	Amount of citric acid present	Numeric	0.00 to 1.00	No missing values	Predictor
Residual Sugar	Sugar remaining after fermentation (g/dm ³)	Numeric	0.9 to 15.5	No missing values	Predictor
Chlorides	Sodium chloride concentration (g/dm ³)	Numeric	0.012 to 0.611	No missing values	Predictor
Free Sulfur Dioxide	Free SO ₂ concentration (mg/dm ³)	Numeric	1 to 72	No missing values	Predictor
Total Sulfur Dioxide	Total SO ₂ concentration (mg/dm ³)	Numeric	6 to 289	No missing values	Predictor
Density	Density of the wine	Numeric	0.9901 to 1.0037	No missing values	Predictor

Variable Name	Definition	Data Type	Allowed Values / Range	Missing Value Handling	Notes
pH	Acidity or alkalinity level	Numeric	2.74 to 4.01	No missing values	Predictor
Sulphates	Potassium sulphate concentration	Numeric	0.33 to 2.00	No missing values	Predictor
Alcohol	Alcohol content (% by volume)	Numeric	8.4 to 14.9	No missing values	Predictor
Wine Quality	Sensory quality score assigned by experts	Integer	3 to 8	No missing values	Target Variable

Table 8

Features Used for Model Development

Component	Description
Predictor Variables	Fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH, sulphates, and alcohol
Target Variable	Wine quality
Feature Engineering	Not performed, as the original laboratory measured variables adequately represented the wine characteristics
Feature Selection	All 11 predictor variables were retained following feature analysis

Component	Description
Data Scaling	Applied only where required for feature analysis (e.g., PCA); tree based models were trained using the original feature values

Appendix B

Table 12

Mapping of Research Questions to Research Hypotheses

Research Question	Null Hypothesis (H_0)	Alternative Hypothesis (H_1)
RQ1	Physicochemical properties do not influence wine quality prediction.	Physicochemical properties influence wine quality prediction.

Research Question	Null Hypothesis (H₀)	Alternative Hypothesis (H₁)
RQ2	Machine learning regression models cannot effectively predict wine quality using physicochemical properties.	Machine learning regression models can effectively predict wine quality using physicochemical properties.
RQ3	There is no meaningful difference in the predictive performance of the selected regression models.	At least one regression model demonstrates better predictive performance than the others.
RQ4	All physicochemical properties contribute equally to wine quality prediction.	Certain physicochemical properties contribute more to wine quality prediction than others.